Introduction to amplicon sequence variant (ASV) analysis

Outline

- Microbial community analysis using amplicon sequence variants (ASVs)
  - What are ASVs
  - What are the differences compared to OTUs
- DADA2 workflow

Amplicon sequence variants

- Biological sequences which can differ by just 1 nucleotide
- Provide increased resolution and sensitivity if compared to OTUs
  - OTU approach clusters sequences together, typically at 97% similarity threshold, in order to remove sequencing errors
  - ASV approach uses abundance and error model in order remove sequencing errors
- Generated without a reference → no reference bias
- Exact sequences, so ASV tables can be compared across studies
- Chimera detection is simpler than with OTUs
- A given target gene should always generate the same ASV
  - Thus, ASVs can be added to reference databases and merged to other datasets

OTU vs ASVs

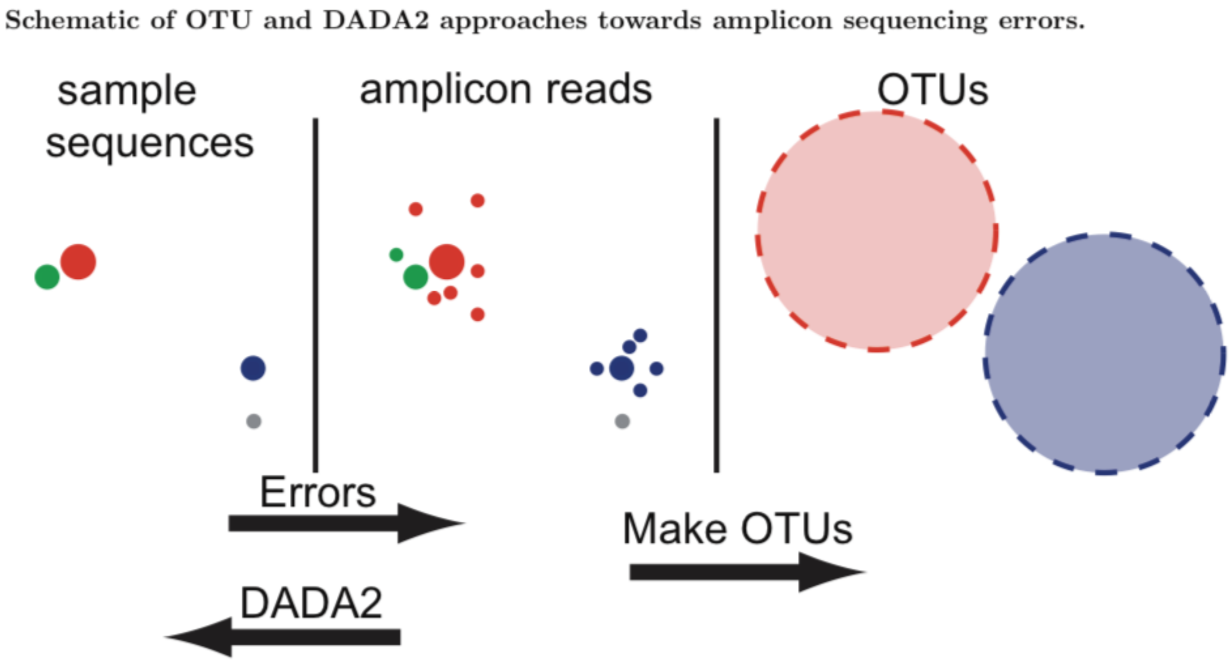


Figure describing the difference between *Amplicon Sequence Variants* (**ASVs**) *vs Operational Taxonomic Units* (**OTUs**) (figure adapted from (Callahan et al. 2016. DADA2: High-Resolution Sample Inference from Illumina Amplicon Data)).

DADA2

- DADA2 = Divisive Amplicon Denoising Algorithm
    - R based library
    - Callahan, B. *et al. 2016.* DADA2: High-resolution sample inference from Illumina amplicon data. https://doi.org/10.1038/nmeth.3869
- Uses the abundance data and the error model to remove sequencing errors and to detect ASVs
- Balance between sensitivity and specificity

DADA2 in Chipster

- The DADA2 analysis tools in Chipster cover the DADA2 tutorial workflow
    - https://benjjneb.github.io/dada2/tutorial.html
- Analysis examples use the MiSeq 16S data (https://mothur.org/wiki/miseq_sop/)
- Analysis of ITS data and data produced with IonTorrent are possible as well
    - Key differences in the workflow will be covered
- Requirements
    - The samples have to be demultiplexed: Split to per-sample FASTQ files

Overview of the ASV preprocessing pipeline in Chipster

1. Make a Tar package
2. Quality control with MultiQC
3. Remove primers/adapters with Cutadapt
4. Filter and trim reads
5. Sample/ASV inference
6. Combine paired reads to contigs
7. Make an ASV table and remove chimeras
8. Assign taxonomy
9. Make a phyloseq object
10. Merge phenodata to the phyloseq object
11. Continue community analysis with the same tools as after the OTU based workflow

Quality control of raw reads

ASV

Quality control of raw reads

Outline

- Different types of quality problems
- FASTQ file format
- Tools for checking read quality
- Tools for improving read quality

What and why?

- Potential problems
  - low confidence bases, Ns
  - adapters
  - …

- Knowing about potential problems in your data allows you to
  - correct for them before you spend a lot of time on analysis
  - take them into account when interpreting results

FASTQ file format

- Four lines per read:

  @read name

  GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTTGTTCAACTCACAGTTT

  + read name

  !''*((((***+))%%%++)(%%%%).1***-+*''))**55CCF>>>>>>CCCCCCC65

- http://en.wikipedia.org/wiki/FASTQ_format
- Do **not** unzip FASTQ files, Chipster can cope with .gz files

Base qualities

- If the quality of a base is 20, the probability that it is wrong is 0.01.
  - Phred quality score Q = -10 * log10 (probability that the base is wrong)

T C A G T A C T C G

40 40 40 40 40 40 40 40 37 35

- Sanger encoding: numbers are shown as ASCII characters so that 33 is added to the Phred score
  - E.g. 39 is encoded as H, the 72nd ASCII character (39+33 = 72)
  - Note that older Illumina data uses different encoding

Base quality encoding systems

```
  SSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSS....................................

  ..LLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLL...................................
  !"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklm
  |                         |    |         |                               |
 33                        59   64        73                              104
  0........................26...31.......40

  0.2......................26...31........41

 S - Sanger        Phred+33,  raw reads typically (0, 40)

 L - Illumina 1.8+ Phred+33,  raw reads typically (0, 41)
```

http://en.wikipedia.org/wiki/FASTQ_format

Tools for checking sequence quality

- **Read quality with MultiQC for many FASTQ files**
  - runs FastQC for all the FASTQ files simultaneously
  - checks base quality and composition, duplication, Ns, k-mers, adapters,…
  - takes a tar package of all the FASTQ files as an input file
- Statistics for primers and adapters with TagCleaner
  - Given an adapter or primer sequence, checks how many reads have it (allowing

tag.statistics.tsv •••

Spreadsheet Text Details

Showing all 9 rows.

#Param	Number_of_Mismatches_or_Splits	Number_of_Sequences	Percentage	Percentage_Sum
tag5	0	54996	95.61	95.61
tag5	1	2114	3.68	99.29
tag5	2	260	0.45	99.74
tag5	3	81	0.14	99.88
tag5	4	36	0.06	99.94
tag5	5	21	0.04	99.98
tag5	6	7	0.01	99.99

Making a Tar package of FASTQ files

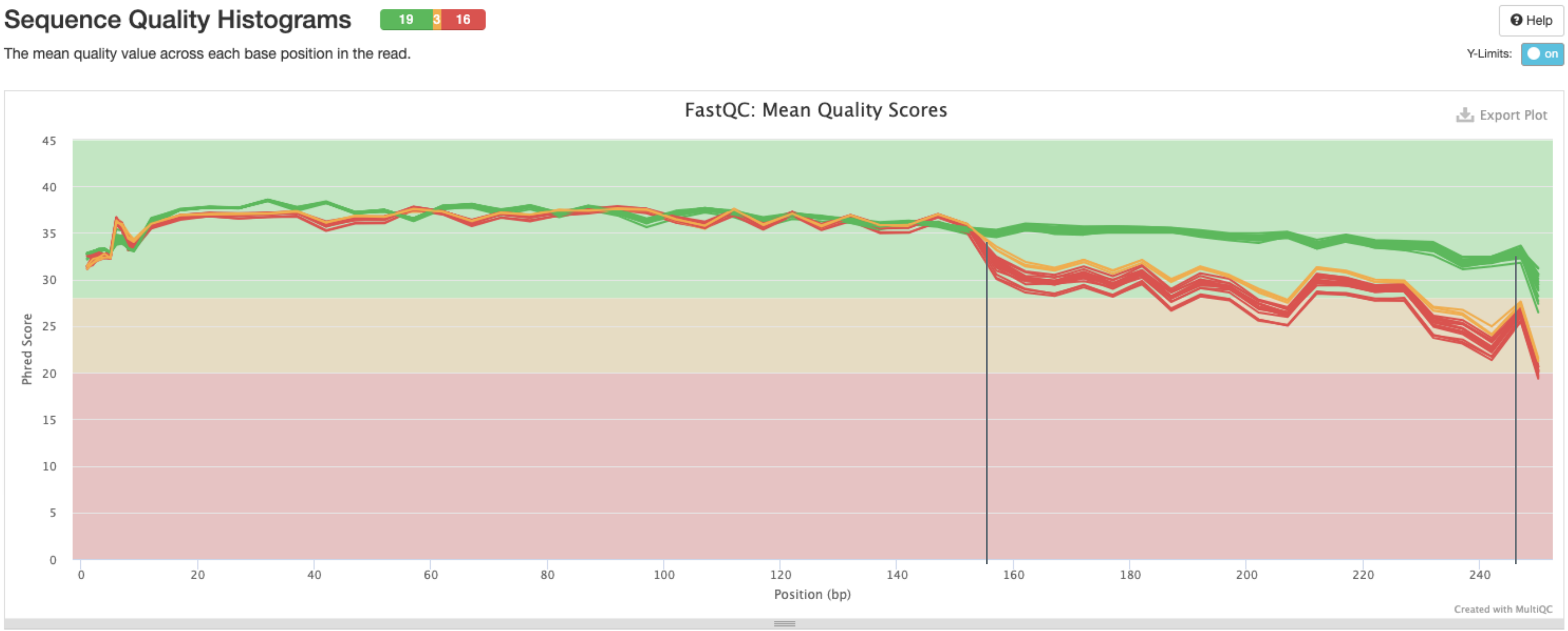
- Use the tool **Utilities / Make Tar package**
- When your Tar package is ready, you can delete the original FASTQ files
    - If you want to look at the individual FASTQ files later, you can always open the Tar package using the tool Utilities / Extract .tar.gz file

MultiQC features

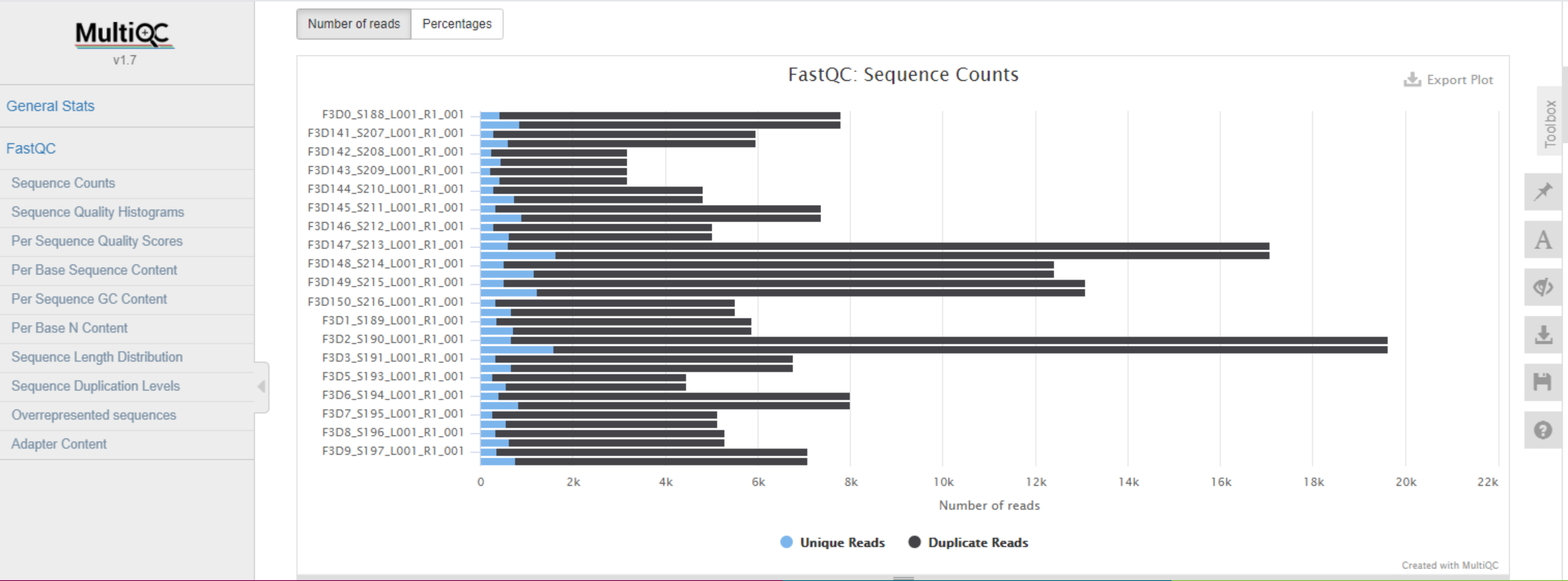
- Interactive plots
- Plots allow you to view the number or percentage of reads
- Traffic lights (they might not be suitable for your data!)
- Toolbox (click on the right side panel) allows you to
    - Highlight samples
    - Show only selected samples
    - Download plots
    - Rename samples
- Good tutorial video https://www.youtube.com/watch?v=qPbIlO_KWNo

Use MultiQC report to decide on trimming parameters

- Quality scores tend to decrease towards the end of the reads
- The quality of reverse reads is often worse in Illumina sequencing



Sequence counts (MultiQC)



Trim primers and adapters with Cutadapt

Outline

- What are adapter sequences
- How to identify adapters and check the correct orientation
- What are the adapter trimming options in Chipster
- How to use Cutadapt to trim adapters
- Things to take into account
- What are the resulting output files

Adapter / primer sequences

- Adapter sequences need to be removed prior to analysis
- Removal is more complicated if some reads extend into the opposite primer
    - Can occur if the amplified region is shorter than the read length
    - If paired end reads, each read may or may not have the forward and reverse primer

Image from DADA2 tutorial,
https://benjjneb.github.io/dada2/ITS_workflow.html

Tools to remove and identify adapters in Chipster

- Cutadapt
- Trimmomatic
  - Can be used to trim adapters from paired end and single end reads
  - Has many quality trimming and filtering options
- Identify primers and the correct orientation
  - Can be used to check if reads contain adapter sequences and in which orientation
- Predict primers/adapters with TagCleaner
  - Can be used to find adapter sequences if those are not known.
- Many trimming tools have an option to trim a fixed length of bases from the start or end of the reads

Identify primers and the correct orientation

- Checks if given 5′ and 3′ primer/adapter sequences are present in the reads and in which orientation
  - Run before and after an adapter trimming tool to see if the adapters were removed
- Remove ambiguous nucleotides first with the tool ”Filter and trim reads with DADA2”
- Test every orientation of the 3′ and 5′ adapters
  - Forward – Compelement – Reverse – Reverse complement
- Input
  - FASTQ files of one sample
  - Tar package of FASTQ files

Output files

Primer_summary.tsv

Showing all 4 rows.

	Forward	Complement	Reverse	RevComp
Forward_reads_5adapter	4214	0	0	0
Forward_reads_3adapter	0	0	0	3590
Reverse_reads_5adapter	0	0	0	3743
Reverse_reads_3adapter	4200	0	0	0

Orientations_summary.txt

```
Check all orientations of the given primers/adapters.

Reads are paired end.
The forward file used to check the adapters: SRR5314314_F_filt.fastq.gz
The reverse file used to check the adapters: SRR5314314_R_filt.fastq.gz

All orientations of the 5' adapter:
Forward:   ACCTGCGGARGGATCA
Complement:   TGGACGCCTYCCTAGT
Reverse:   ACTAGGRAGGCGTCCA
RevComp:   TGATCCYTCCGCAGGT


All orientations of the 3' adapter:
Forward:   GAGATCCRTTGYTRAAAGTT
Complement:   CTCTAGGYAACRAYTTTCAA
Reverse:   TTGAAARTYGTTRCCTAGAG
RevComp:   AACTTTYARCAAYGGATCTC
```

Trim adapters with Cutadapt

- Based on Cutadapt
  - MARTIN, Marcel. 2011. Cutadapt removes adapter sequences from high-throughput sequencing reads. https://doi.org/10.14806/ej.17.1.200.
- Input
  - Tar package of FASTQ files
- Need to specify whether the reads are paired or single end
- Give the 5′ and 3′ adapters in the orientation they are found in the forward sequence
  - Use the tool "Identify primers and the correct orientation" to find and copy the right orientation
- 3′ adapters and any sequence following it is removed
- 5′ adapter and any sequence preceding it is removed
- Be carefull not to trim adapters from wrong ends!

Removing primers with Cutadapt

Remove primers and adapters with Cutadapt

Parameters

Reset All

Are the reads paired end or single end — paired

If your reads are paired end, the reverse complement of the 3' and 5' adapters will be removed from the reverse reads.

The 5' adapter: — GTCAGCTCGTGYYGTGAG

Give here the 5 end adapter/primer.

The 3' adapter: — TTGYACACACCGCCCGT

Give here the 3 end adapter/primer in the orientation it exists in the forward/single reads, usually in reverse complement.

reverse complement

Remove reads which were not trimmed — yes

Remove reads which did not contain an adapter.

Input files

Tar package containing the FASTQ files — zippedFastq.tar

Example

1)

Showing all 4 rows.

	Forward	Complement	Reverse	RevComp
Forward_reads_5adapter	4214	0	0	0
Forward_reads_3adapter	0	0	0	3590
Reverse_reads_5adapter	0	0	0	3743
Reverse_reads_3adapter	4200	0	0	0

2)

```
All orientations of the 5' adapter:
Forward:   ACCTGCGGARGGATCA
Complement:   TGGACGCCTYCCTAGT
Reverse:   ACTAGGRAGGCGTCCA
RevComp:   TGATCCYTCCGCAGGT


All orientations of the 3' adapter:
Forward:   GAGATCCRTTGYTRAAAGTT
Complement:   CTCTAGGYAACRAYTTTCAA
Reverse:   TTGAAARTYGTTRCCTAGAG
RevComp:   AACTTTYARCAAYGGATCTC
```

3)

Parameters

Reset All

Is the data paired end or single end reads Are all the reads paired end, so one forward and one reverse FASTQ file for one sample.	paired
The 5' adapter: Give here the 5 end adapter/primer	ACCTGCGGARGGATCA
The 3' adapter: Give here the 3 end adapter/primer	AACTTTYARCAAYGGATCTC

Output files

- Adapters_removed.tar
- Report.txt
- Samples.fastqs.txt

Samples.fastqs.txt

SRR5314314	SRR5314314_1.fastq.gz	SRR5314314_2.fastq.gz
SRR5314315	SRR5314315_1.fastq.gz	SRR5314315_2.fastq.gz
SRR5314316	SRR5314316_1.fastq.gz	SRR5314316_2.fastq.gz
SRR5314317	SRR5314317_1.fastq.gz	SRR5314317_2.fastq.gz
SRR5314331	SRR5314331_1.fastq.gz	SRR5314331_2.fastq.gz
SRR5314332	SRR5314332_1.fastq.gz	SRR5314332_2.fastq.gz
SRR5314333	SRR5314333_1.fastq.gz	SRR5314333_2.fastq.gz
SRR5314334	SRR5314334_1.fastq.gz	SRR5314334_2.fastq.gz
SRR5314335	SRR5314335_1.fastq.gz	SRR5314335_2.fastq.gz

Report.txt

- Big file, contains a report for every sample
- Contains all the relevant information how the adapters were removed
- Always check where the adapters were found and how many bases were removed

```
This is cutadapt 4.1 with Python 3.8.11
Command line parameters: -g ACCTGCGGARGGATCA -a AACTTTYARCAAYGGATCTC -G GAGATCCRTTGYTRAAAGTT -A TGATCCYTCCGCAGGT -n 2 -j 2 -o out
Processing paired-end reads on 2 cores ...
Finished in 2.77 s (44 µs/read; 1.37 M reads/minute).

=== Summary ===

Total read pairs processed:            63,398
  Read 1 with adapter:                 47,992 (75.7%)
  Read 2 with adapter:                 48,032 (75.8%)
Pairs written (passing filters):       63,398 (100.0%)

Total basepairs processed:    31,737,011 bp
  Read 1:    15,867,972 bp
  Read 2:    15,869,039 bp
Total written (filtered):     22,831,611 bp (71.9%)
  Read 1:    11,374,379 bp
  Read 2:    11,457,232 bp

=== First read: Adapter 1 ===

Sequence: ACCTGCGGARGGATCA; Type: regular 5'; Length: 16; Trimmed: 46986 times

Minimum overlap: 3
No. of allowed errors:
1-9 bp: 0; 10-16 bp: 1

Overview of removed sequences
length  count   expect  max.err error counts
3       4       990.6   0       4
58      2       0.0     1       1 1
59      3       0.0     1       1 2
60      6       0.0     1       3 3
61      10      0.0     1       7 3
62      15      0.0     1       9 6
63      51      0.0     1       18 33
64      550     0.0     1       224 326
65      46191   0.0     1       18648 27543
66      143     0.0     1       60 83
67      4       0.0     1       1 3
68      3       0.0     1       0 3
70      1       0.0     1       0 1
71      1       0.0     1       0 1
73      1       0.0     1       0 1
94      1       0.0     1       1


=== First read: Adapter 2 ===

Sequence: AACTTTYARCAAYGGATCTC; Type: regular 3'; Length: 20; Trimmed: 45969 times

Minimum overlap: 3
No. of allowed errors:
1-9 bp: 0; 10-19 bp: 1; 20 bp: 2
```

Run “Identify primers and the correct orientation” again

- That looks great, we managed to remove the adapters

Showing all 4 rows.

	Forward	Complement	Reverse	RevComp
Forward_reads_5adapter	0	0	0	0
Forward_reads_3adapter	0	0	0	0
Reverse_reads_5adapter	0	0	0	0
Reverse_reads_3adapter	0	0	0	0

- If we didn’t remove those sequences with ‘N’ bases, the result would likely look like this:

Showing all 4 rows.

	Forward	Complement	Reverse	RevComp
Forward_reads_5adapter	1060	1060	1060	1060
Forward_reads_3adapter	1060	1060	1060	1060
Reverse_reads_5adapter	1023	1023	1023	1023
Reverse_reads_3adapter	1023	1023	1023	1023

Filter and trim reads with DADA2

Outline

- Why read filtering and trimming is important
- Things to take into account
- What trimming and filtering options are available
- How to set the parameters in Chipster
- What are the output files

Filter and trim reads with DADA2

- Critical step in ASV workflow
  - Ns need to be removed
  - Remove low quality ends and too short reads
  - Cleaned data runs more efficiently and gives more accurate results
  - Reads matching the phiX genome are removed
- Based on the DADA2 tool filterAndTrim()
- Optimal parameter values depend on your data
  - Use MultiQC report to decide on parameter values
- If paired end reads, both reads need to pass the filter in order to be kept
- Input file: Tar package of FASTQ files
  - You can make it with the tool "Make a tar package"
  - FASTQ files can be compressed

Filter based on the number of Ns and expected errors

- Ns need to be removed before the next analysis steps
- Maximun number of expected errors allowed for a read
  - Expected errors are calculated as the sum of error probabilities, $EE = \sum 10^{-Q/10}$
  - More info https://doi.org/10.1093/bioinformatics/btv401
  - Default is 2

Discard input sequences with more than specified number of Ns

0

Sequences with more than the specified number of Ns will be discarded. Note that the dada function does not allow any Ns.

Discard forward sequences with more than the specified number of expected errors

After truncation, reads with more than this amount of expected errors will be discarded. If this parameter is not set, no expected error filtering is done. You can use this parameter for single and paired end reads.

Discard reverse sequences with more than the specified number of expected errors

After truncation, reads with more than this amount of expected errors will be discarded. If this parameter is not set, no expected error filtering is done. Use only for paired end reads.

Output files

- Filtered.fastqs.tar
- Summary.tsv
- Samples.fastqs.txt

Summary.tsv

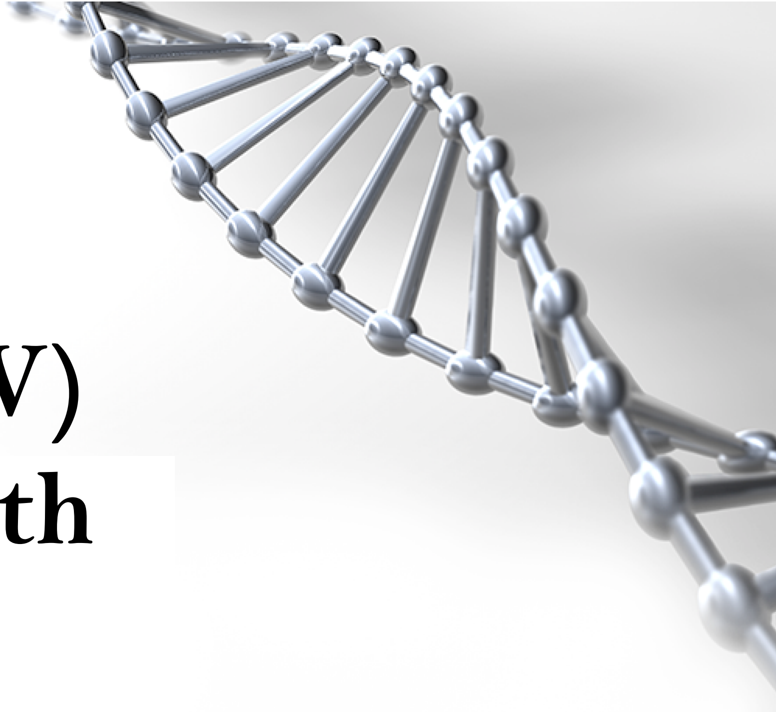
Showing all 19 rows.

	reads.in	reads.out
F3D0	7793	7135
F3D141	5958	5477
F3D142	3183	2928
F3D143	3178	2959
F3D144	4827	4342
F3D145	7377	6787
F3D146	5021	4580
F3D147	17070	15719
F3D148	12405	11452
F3D149	13083	12059
F3D150	5509	5046
F3D1	5869	5316
F3D2	19620	18137
F3D3	6758	6282
F3D5	4448	4069
F3D6	7989	7394
F3D7	5129	4777
F3D8	5294	4885
F3D9	7070	6520

Samples.fastqs.txt

F3D0	F3D0_S188_L001_R1_001.fastq	F3D0_S188_L001_R2_001.fastq
F3D141	F3D141_S207_L001_R1_001.fastq	F3D141_S207_L001_R2_001.fastq
F3D142	F3D142_S208_L001_R1_001.fastq	F3D142_S208_L001_R2_001.fastq
F3D143	F3D143_S209_L001_R1_001.fastq	F3D143_S209_L001_R2_001.fastq
F3D144	F3D144_S210_L001_R1_001.fastq	F3D144_S210_L001_R2_001.fastq
F3D145	F3D145_S211_L001_R1_001.fastq	F3D145_S211_L001_R2_001.fastq
F3D146	F3D146_S212_L001_R1_001.fastq	F3D146_S212_L001_R2_001.fastq
F3D147	F3D147_S213_L001_R1_001.fastq	F3D147_S213_L001_R2_001.fastq
F3D148	F3D148_S214_L001_R1_001.fastq	F3D148_S214_L001_R2_001.fastq
F3D149	F3D149_S215_L001_R1_001.fastq	F3D149_S215_L001_R2_001.fastq
F3D150	F3D150_S216_L001_R1_001.fastq	F3D150_S216_L001_R2_001.fastq

Sample (ASV) inference with DADA2



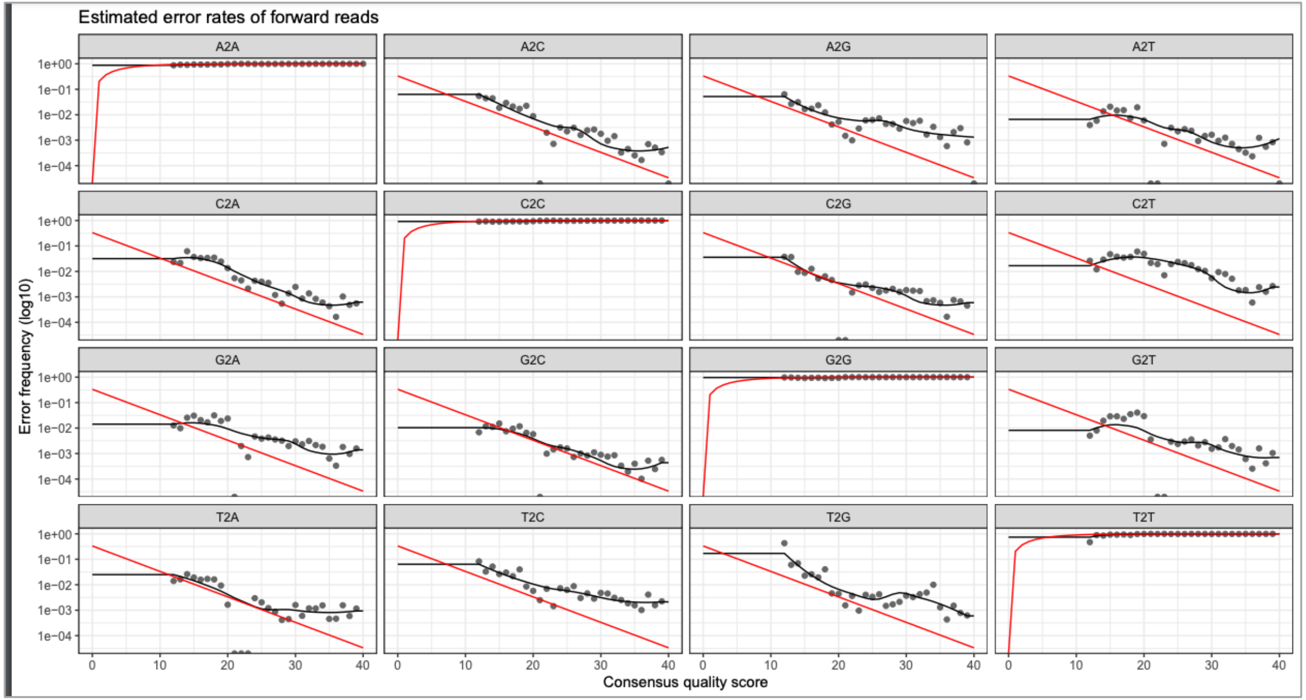
ASV inference

- Separating biological variation and sequencing errors
    - run-specific sequence quality scores
    - read abundances
- Biological sequences are more likely to be repeatedly observed than are error-containing sequences
- Two steps:
    - error model creation
    - identify ”real” ASVs using the error model and abundance information
        - divisive hierarchical clustering algorithm

Considerations for ASV inference

- Samples should be in separate FASTQ files (demultiplexed)
- Primers and sequencing adapters removed
- No ambiguous bases (N) (⟶ filter and trim)
- Error rates are specific to sequencing runs
  - multiple runs ⟶ carry out ASV inference separately for each sequencing run and combine ASV tables (⟶ Chipster tool Merge two ASV tables)

Visualize estimated error rates



- Points = observed error rates
- Black line = estimated error rates after convergence
- Red line = Expected error rates based on the definition of the Q-scores

Input in Chipster

- Tar package of FASTQ files containing filtered reads
  - Specify if reads are paired end or single end and if they were produced with IonTorrent
  - Independent or pseudo-pooling
  - Forward and reverse reads are processed independently

Parameter options - pooling

- **Independent**
  - ASVs are inferred individually from each sample
  - Computationally easiest way
- **Pseudo-pooling**
  - Can increase sensitivity: might find rare variants
  - Runs the individual processing twice. Uses the ASVs found in all samples in the first run as priors in the second run.
  - Computationally harder

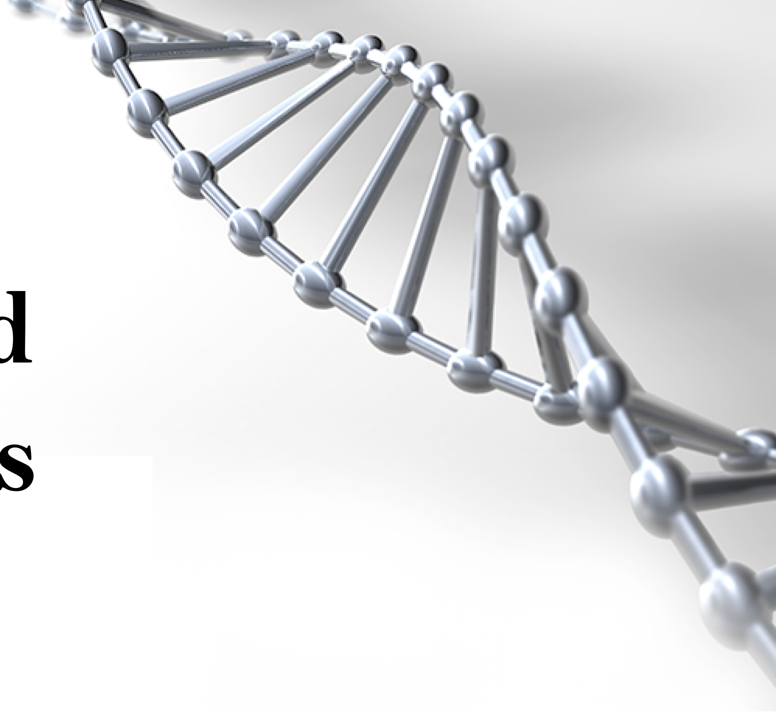
Parameter options - pooling

- **Independent**
  - ASVs are inferred individually from each sample
  - Computationally easiest way
- **Pseudo-pooling**
  - Can increase sensitivity: might find rare variants
  - Runs the individual processing twice. Uses the ASVs found in all samples in the first run as priors in the second run.
  - Computationally harder
- The number of different ASVs found from all the samples is the same
  - The abundance of different ASVs in each sample can increase with pseudo-pooling
- The best choice depends on the data
- Note that singletons are not detected
  - Sequences with an abundance of 1 can't form a new cluster

Output

- DADA class objects saved as .Rda objects
  - Forward and reverse objects in separate files
- Summary.txt → Information on learnErrors() and dada() functions
  - Key parameters used
  - Number of unique sequences in each file
  - Number of ASVs inferred from each file

```
$F3D0_F_filt.fastq.gz
128 sequence variants were inferred from 1979 input unique sequences.

$F3D141_F_filt.fastq.gz
97 sequence variants were inferred from 1477 input unique sequences.
```



Combine paired reads to contigs with DADA2

Outline

- How paired reads are merged to contigs with DADA2
- What are the parameter options
- What are the resulting output files

Combine paired reads to contigs with DADA2

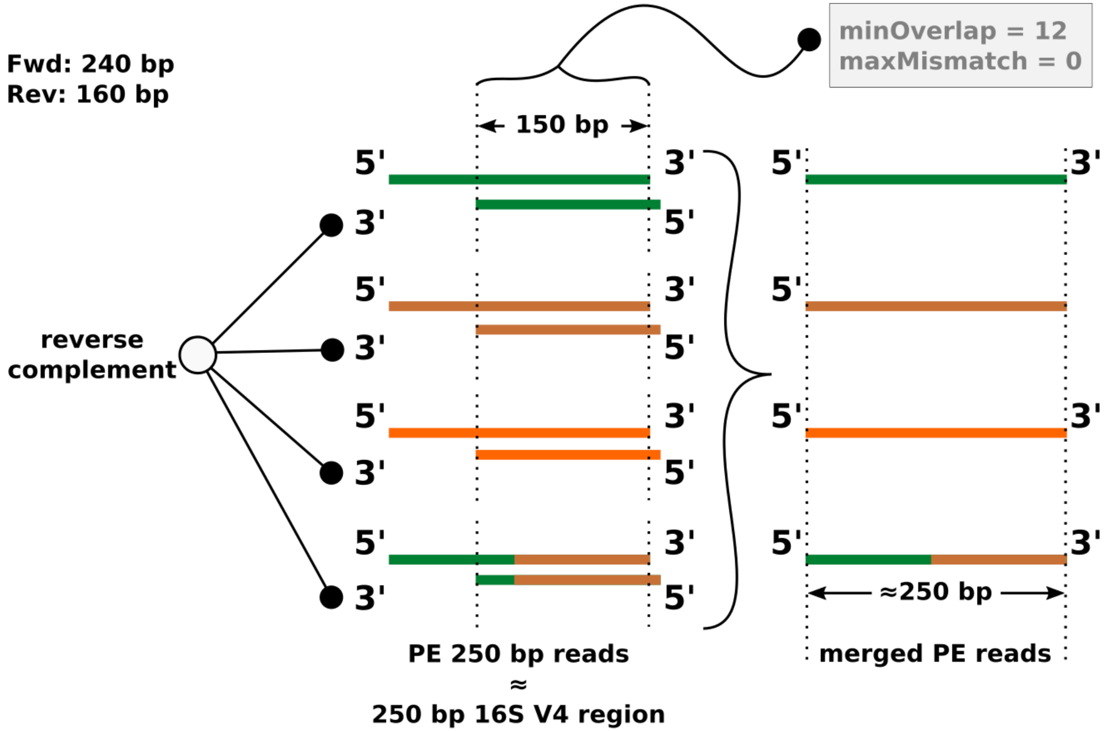
- Tries to merge each denoised pair of forward and reverse reads
  - Aligns forward read with the reverse compelement of the reverse read
- Input files
  - Tar package containing all the filtered forward and reverse FASTQ files
  - two denoised dada-class objects from the ”Sample inference” tool
    - forward_dada.Rda
    - reverse_dada.Rda
  - Make sure that the input files are assigned correctly

Parameter options

By default:

- Overlap region needs to be at least 12 base pairs long
- No mismatches are allowed in the overlap region
- Overhangs are not trimmed off
  - Can result when the reverse reads extend past the start of the forward reads and vice versa

Example: Paired end Illumina data



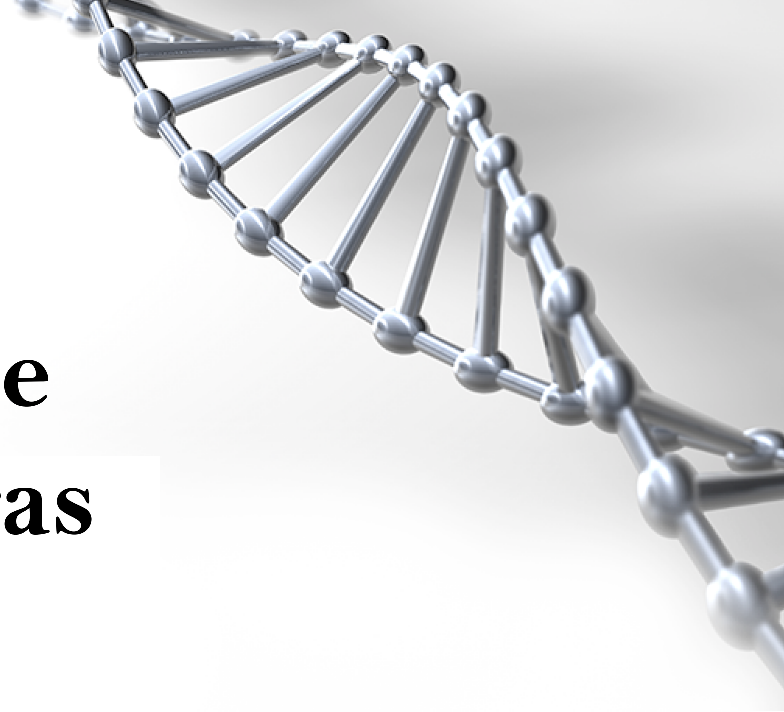
merge denoised forward and reverse reads - *mergePairs()*

Image by Antonio Sousa, 16S rRNA gene amplicon - upstream data analysis, https://igcbioinformatics.github.io/biomeshinycourse/pages/dada2/Biodata.ptCrashCourses.html

Output files

1. Object contigs.Rda containing a list of data frames
2. Summary table contigs_summary.tsv
   - Per sample: how many forward reads, reverse reads and contigs
     - Reads that don’t overlap or have a pair are discarded

	Forward dada object	Reverse dada object	After make contigs
F3D0	6976	6978	6533
F3D141	5331	5350	4973
F3D142	2799	2832	2595
F3D143	2822	2867	2552
F3D144	4151	4224	3643
F3D145	6592	6628	6079
F3D146	4447	4470	3968



Make an ASV table and remove chimeras with DADA2

Outline

- How to create an amplicon sequence variant table
- How does it look like
- What are chimeras and how to remove them
- What are the parameter options
- What are the resulting output files

Amplicon sequence variant table

- Input file can be either:
  1. "contigs.Rda" - paired end reads merged to contigs
  2. "dada_forward.Rda" – single-end reads
- Shows the distribution of ASVs in each sample

	ASV1	ASV2	ASV3	ASV4	ASV5	ASV6	ASV7	ASV8	ASV9	ASV10	ASV11	ASV12	ASV13
F3D0	579	345	449	430	154	470	282	184	45	158	17	217	52
F3D141	444	362	345	502	189	331	243	321	167	130	168	146	12
F3D142	289	304	158	164	180	181	163	89	89	78	42	98	103
F3D143	228	176	204	231	130	244	152	83	109	67	78	111	43
F3D144	421	277	302	357	104	353	240	41	158	155	269	146	16
F3D145	645	489	522	583	307	476	396	125	202	229	317	258	22
F3D146	325	230	254	388	179	275	214	71	113	88	178	147	4
F3D147	1495	1215	913	1089	453	1182	861	75	769	269	448	560	147
F3D148	863	729	581	853	443	872	579	507	409	198	411	432	18
F3D149	883	779	723	897	417	637	560	515	426	288	478	301	88
F3D150	317	229	399	471	169	216	238	120	241	149	61	98	64
F3D1	405	353	231	69	140	41	96	190	69	106	102	40	129
F3D2	3488	1587	1175	472	338	115	325	1211	434	609	55	41	330

samples in rows
ASVs in columns

Remove chimeras

- Chimeras are artifact sequences formed by two or more biological sequences
    - Incomplete amplification during PCR allows subsequent PCR cycles to use a partially extended strand to bind to the template of a similar sequence
    - The partially extended strand then acts as a primer to extend and form a chimeric sequence
- Based on removeBimeraDenovo() of DADA2 library
- Used to identify and remove chimeras from the ASV table

Parameter options

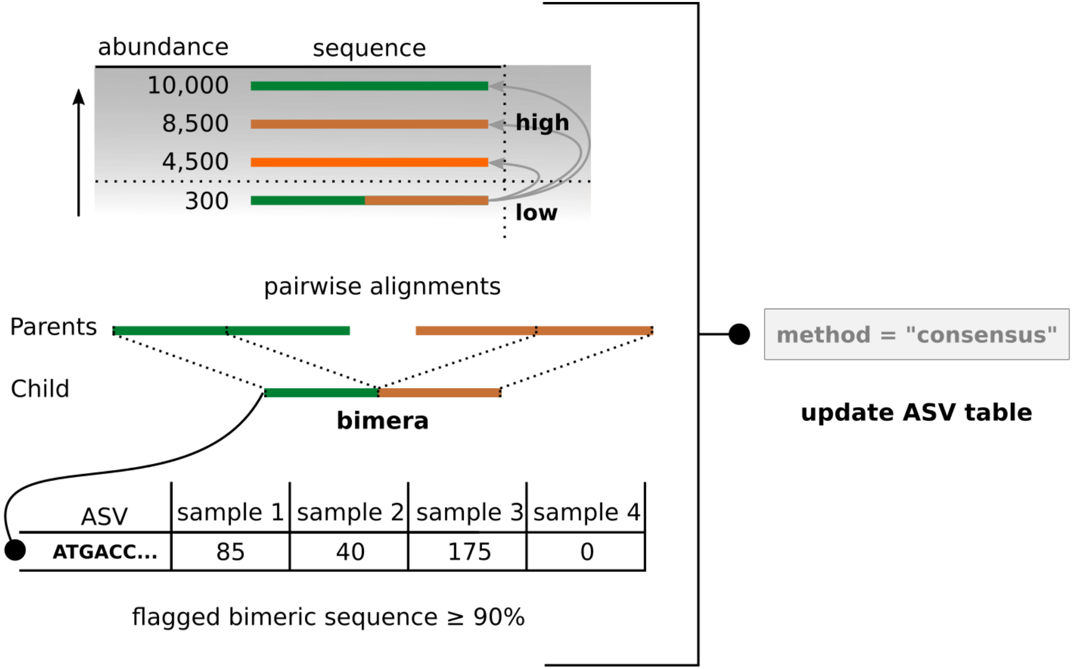
- **Consensus**
  - Chimeras are identified on sample-by-sample basis
  - If an ASV is identified as a chimera in most of the samples, it will be removed
- **Pooled**
  - Each sequence is compared against the more abundant sequences of all samples

How it works

- Assumes that chimeras arise from two parent sequences
  - Bimera is a chimera formed by exactly two parent sequences

- Performs Needleman-Wunsch pairwise alignments to compare less abundant sequences with the most abundant sequences

- If the less abundant sequence can be reconstructed by combining a left segment and a right segment from two more abundant sequences, the sequence is removed as chimeric



remove chimeras - *removeBimeraDenovo()*

Image by Antonio Sousa, 16S rRNA gene amplicon - upstream data analysis, https://igcbioinformatics.github.io/biomeshinycourse/pages/dada2/Biodata.ptCrashCourses.html

Output files

1. Seqtab_nochim.Rda
2. Sequence_table_nochim.tsv
3. Summary.txt
4. Reads_summary.tsv

sequence_table_nochim.tsv

Showing all 19 rows.

	ASV1	ASV2	ASV3	ASV4	ASV5	ASV6	ASV7	ASV8	ASV9	ASV10	ASV11	ASV12	ASV13	ASV14	ASV15	ASV16	ASV17	ASV18
F3D0	579	345	449	430	154	470	282	184	45	158	17	217	52	104	93	80	100	69
F3D141	444	362	345	502	189	331	243	321	167	130	168	146	12	65	33	103	149	43
F3D142	289	304	158	164	180	181	163	89	89	78	42	98	103	64	12	52	116	30
F3D143	228	176	204	231	130	244	152	83	109	67	78	111	43	61	9	40	0	20
F3D144	421	277	302	357	104	353	240	41	158	155	269	146	16	81	11	113	0	45
F3D145	645	489	522	583	307	476	396	125	202	229	317	258	22	125	15	126	195	105
F3D146	325	230	254	388	179	275	214	71	113	88	178	147	4	58	25	35	0	35
F3D147	1495	1215	913	1089	453	1182	861	75	769	269	448	560	147	292	74	306	260	143
F3D148	863	729	581	853	443	872	579	507	409	198	411	432	18	199	56	270	259	117
F3D149	883	779	723	897	417	637	560	515	426	288	478	301	88	164	42	177	82	119
F3D150	317	229	399	471	169	216	238	120	241	149	61	98	64	75	19	30	0	49
F3D1	405	353	231	69	140	41	96	190	69	106	102	40	129	28	325	0	0	31
F3D2	3488	1587	1175	472	338	115	325	1211	434	609	55	41	330	107	368	17	70	190
F3D3	988	602	465	200	402	25	167	381	307	298	171	0	94	61	46	24	0	103
F3D5	327	268	284	158	151	23	123	207	178	207	61	0	48	38	87	36	57	37
F3D6	1014	674	588	404	476	17	282	261	205	242	45	0	422	106	57	0	0	39
F3D7	648	504	438	314	470	11	195	213	176	276	16	0	117	73	40	0	0	22
F3D8	272	352	349	147	582	0	130	286	113	197	19	0	145	65	45	6	0	22
F3D9	511	423	482	206	596	0	210	438	146	225	27	0	182	72	99	0	0	38

ASVs renamed for visualization purposes, the .Rda object contains still the full DNA sequences

Reads_summary.tsv

Showing all 19 rows.

	After make contigs	Removed chimeras
F3D0	6533	6521
F3D141	4973	4850
F3D142	2595	2521
F3D143	2552	2518
F3D144	3643	3504
F3D145	6079	5820
F3D146	3968	3879
F3D147	14233	13006
F3D148	10528	9935
F3D149	11155	10653
F3D150	4349	4240
F3D1	5028	5017
F3D2	17431	16835
F3D3	5850	5486
F3D5	3716	3716
F3D6	6865	6678
F3D7	4426	4215
F3D8	4560	4531
F3D9	6093	6016

- ❖ Note: it's common that many of the ASVs are removed as chimeric, but most of the reads should not!

Summary.txt

```
After  dada() algorithm sequence table consist of:
19  samples and  279  amplicon sequence variants

Distribution of the amplicon sequence variant's lengths: Column names are the sequence lengths


        251 252 253 254 255
Counts:   1  85 186   5   2

###Removing Chimeras:###
Identified  61  bimeras out of  279  input sequences
Total amount of ASVs is: 218
```

Assign taxonomy with DADA2

Outline

- Methods to assign taxonomy to ASVs
- What are the parameter options
- What things to take into account
- What are the resulting output files

Assign Taxonomy

- Based on the assignTaxonomy() function of DADA2 library
- Uses the naive Bayesian classifier method of Wang to assign taxonomy across multiple ranks
  - Compares the k-mer profile of the query sequences against the reference sequences with assigned taxonomies
  - Calculates the bootstrapping confidence score for the assignment
  - Uses k-mer size of 8 and 100 iterations
- There is a separate command for species level assignment called addSpecies()

Assing Taxonomy – Input files

- Input: ASV table saved as .Rda file
- Uses SILVA (currently v. 138.2) reference files by default
  - To use other reference databases, download the DADA2 supported reference file and bring it to Chipster
  - DADA2 supported reference databases: https://benjjneb.github.io/dada2/training.html

silva

MANUAL > RELEASE 138.2

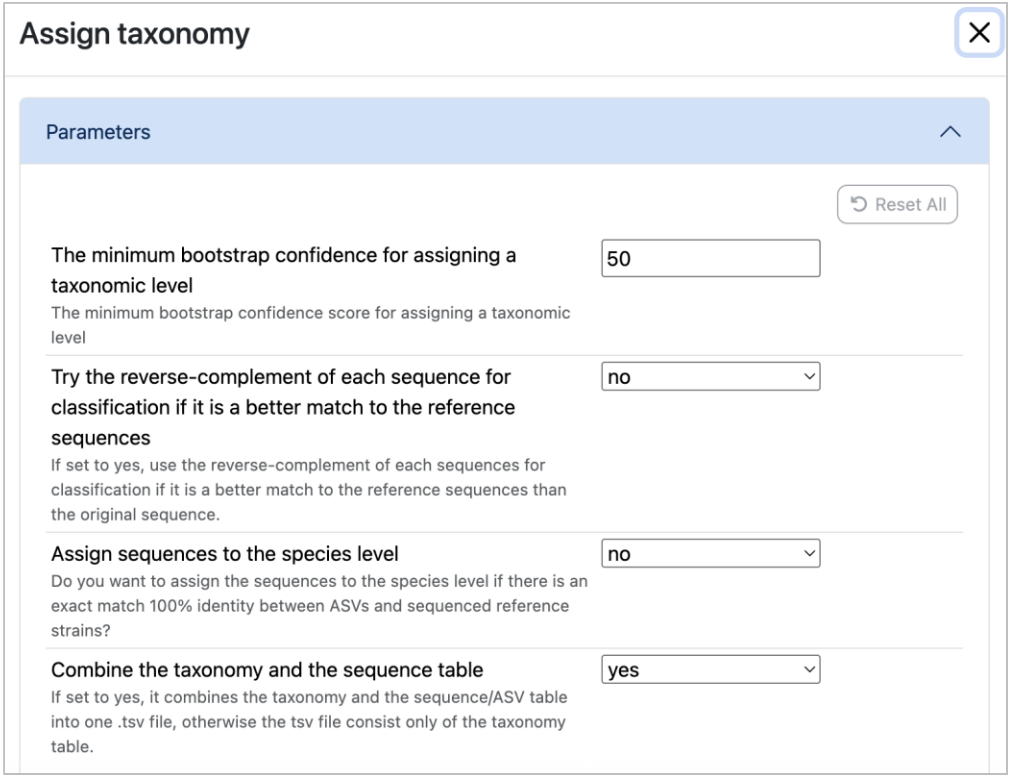
Basic statistics for the SILVA databases

	SSU Ref	SSU Ref NR
Version	138.2	138.2
Total	2,224,690	510,495
Bacteria	1,982,812	431,166
Archaea	69,198	20,389
Eukaryota	172,680	58,940

https://www.arb-silva.de/

Parameter options

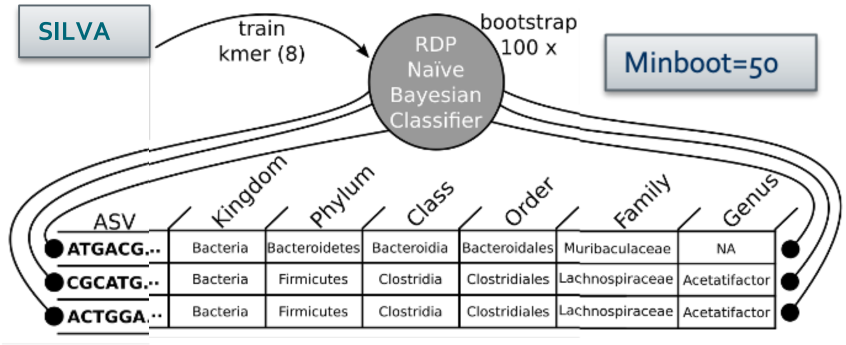
- Specify the minimum bootstrap confidence score to assign taxonomy to ASVs
  - Threshold of 50% means that at least half of the iterations should return the same assignment for each level
  - By default threshold is set to 50% which is recommended for sequences shorter than 250 bases
    - Otherwise 80% is recommended



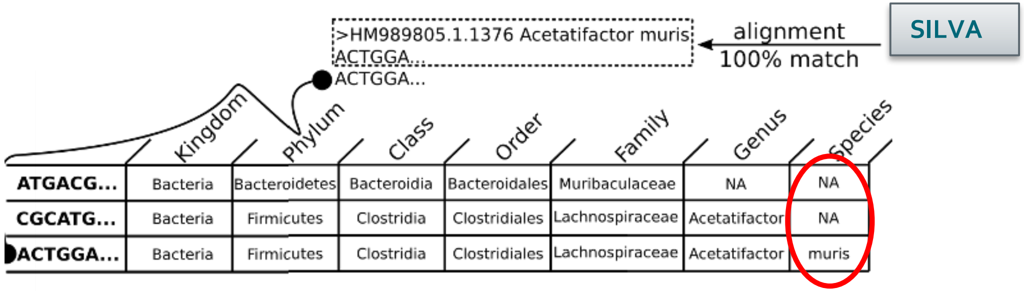
AddSpecies()

- If the parameter "Exact species level assignment " is set to yes, the addSpecies() function for species level assignment is used
- Based on exact (100% identity) string matching against a reference database
  - Assign to species level if there is no ambiquity and all exact matches were to the same species
  - 100% identity matching is recommended for 16S amplicon data
    - See Robert. C. Edgar, (2018), Updating the 97% identity threshold for 16S ribosomal RNA OTUs
- Uses SILVA reference files if those were used for assignTaxonomy() as well
  - Otherwise give your own reference file as input

Example:



AssignTaxonomy()



addSpecies()

Images by Antonio Sousa, 16S rRNA gene amplicon - upstream data analysis,
https://igcbioinformatics.github.io/biomeshinycourse/pages/dada2/Biodata.ptCrashCourses.html

Output files

1. taxonomy-assignment-matrix.Rda
2. taxa_seqtab_combined.tsv
    - If "Combine the taxonomy and the sequence table" is set to yes, both of those tables are combined to taxa_seqtab_combined.tsv
    - The names of the ASVs are renamed for visualization purposes. The Rda object still has the full DNA sequences

Showing the first 100 of 218 rows. View in full screen to see all rows. Full Screen

	Kingdom	Phylum	Class	Order	Family	Genus
ASV1	Bacteria	Bacteroidota	Bacteroidia	Bacteroidales	Muribaculaceae	NA
ASV2	Bacteria	Bacteroidota	Bacteroidia	Bacteroidales	Muribaculaceae	NA
ASV3	Bacteria	Bacteroidota	Bacteroidia	Bacteroidales	Muribaculaceae	NA
ASV4	Bacteria	Bacteroidota	Bacteroidia	Bacteroidales	Muribaculaceae	NA
ASV5	Bacteria	Bacteroidota	Bacteroidia	Bacteroidales	Bacteroidaceae	Bacteroides
ASV6	Bacteria	Bacteroidota	Bacteroidia	Bacteroidales	Muribaculaceae	NA
ASV7	Bacteria	Bacteroidota	Bacteroidia	Bacteroidales	Muribaculaceae	NA
ASV8	Bacteria	Bacteroidota	Bacteroidia	Bacteroidales	Rikenellaceae	Alistipes
ASV9	Bacteria	Bacteroidota	Bacteroidia	Bacteroidales	Muribaculaceae	NA
ASV10	Bacteria	Bacteroidota	Bacteroidia	Bacteroidales	Muribaculaceae	NA

Make a phyloseq object

Outline

- What is a phyloseq object
- How to create it
- What are the resulting output files
- How to extract information from the phyloseq object

Make a phyloseq object

- Phyloseq is an R package to import, store and analyze microbial community data
  - Stores all related sequencing data which makes the analyses easier
- In order to create a phyloseq .Rda object you need to give as input
  1. ASV table saved as .Rda object
  2. Taxonomy table saved as .Rda object

https://joey711.github.io/phyloseq/

Output files

- Creates a phyloseq object called ps_nophe.Rda
  - Produces a phenodata file used to specify sample information
- Summary file : ps_summary.txt
  - The full DNA sequences of ASVs stored to refseq() slot
  - You can get those with the tool “Extract information from the phyloseq object”

```
### Phyloseq object ###



phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 218 taxa and 19 samples ]
tax_table()   Taxonomy Table:    [ 218 taxa by 7 taxonomic ranks ]
refseq()      DNAStringSet:      [ 218 reference sequences ]
```

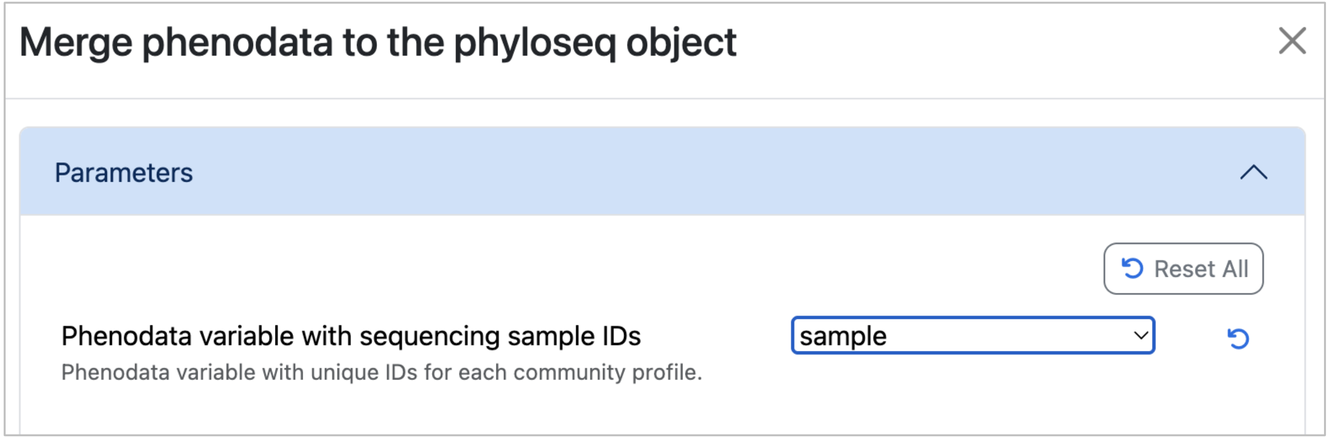
Phenodata table

- Table you can edit via the Chipster interface
- Use to specify and sort samples to different groups
    - Makes data analysis easier

sample	original_name	group ×	diet ×	description
F3D0		a	low	
F3D141		b	low	
F3D142		b	low	
F3D143		b	low	
F3D144		b	low	
F3D145		b	low	
F3D146		b	low	
F3D147		b	low	
F3D148		b	high	
F3D149		b	high	
F3D150		b	high	
F3D1		a	high	
F3D2		a	high	
F3D3		a	high	
F3D5		a	high	
F3D6		a	high	
F3D7		a	high	
F3D8		a	high	
F3D9		a	high	

Merge phenodata to the phyloseq object

- Store the sample information from the Phenodata table to the phyloseq object
- As input, give the phyloseq .Rda object and the Phenodata table
- Specify the column form the Phenodata table which contains the sample names / IDs



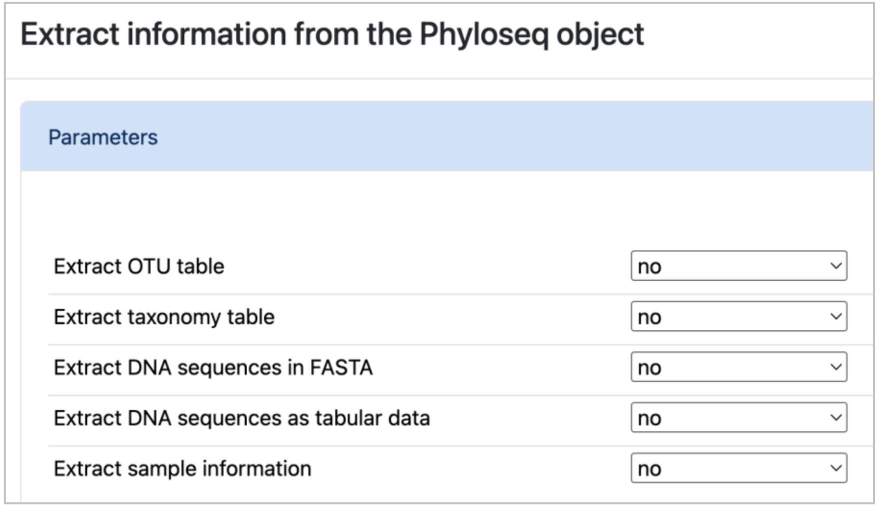
Output files

1. Phyloseq object ps.Rda
2. Summary file ps_sample_summary.txt
   - If you modify the phenodata table, you need to merge the phenodata table to the phyloseq object again!

```
### Phyloseq object with sample information combined###


phyloseq-class experiment-level object
otu_table()   OTU Table:         [ 218 taxa and 19 samples ]
sample_data() Sample Data:       [ 19 samples by 5 sample variables ]
tax_table()   Taxonomy Table:    [ 218 taxa by 7 taxonomic ranks ]
refseq()      DNAStringSet:      [ 218 reference sequences ]


### Sample names ###


 [1] "F3D0"   "F3D141" "F3D142" "F3D143" "F3D144" "F3D145" "F3D146" "F3D147"
 [9] "F3D148" "F3D149" "F3D150" "F3D1"   "F3D2"   "F3D3"   "F3D5"   "F3D6"
[17] "F3D7"   "F3D8"   "F3D9"


### Sample variables ###


[1] "sample"        "original_name" "chiptype"      "group"
[5] "description"
```

Extract information from the phyloseq object

- Used to extract information stored to the phyloseq object
- Give a phyloseq object .Rda as input
  - Can be used to access information after the DADA2 or Mothur based workflow



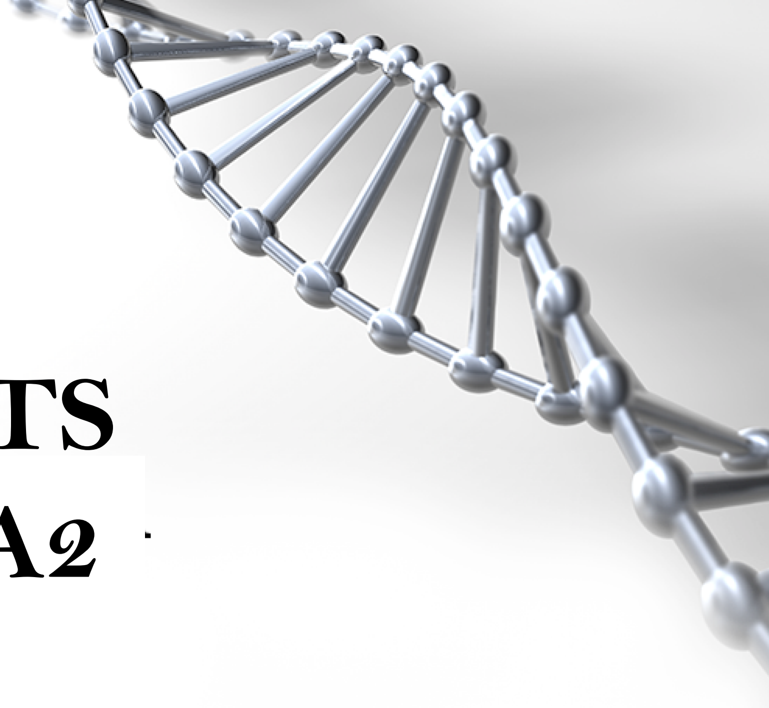
Full DNA sequences stored in the refseq() slot of the Phyloseq object

- Get access to the full sequences of each ASV
    - Can be used to merge the found ASVs with other datasets and index into reference databases

Showing the first 100 of 218 rows. View in full screen to see all rows. Full Screen

ASV1	TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGAAGATCAAGTCAGCGGTAAAATTGAGAGGCTCAACCTCTTCGAGCCGTTGA
ASV2	TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGACTCTCAAGTCAGCGGTCAAATCGCGGGGCTCAACCCCGTTCCGCCGTTGA
ASV3	TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTGTTAAGTCAGCGGTCAAATGTCGGGGCTCAACCCCGGCCTGCCGTTGA
ASV4	TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGCTTTTAAGTCAGCGGTAAAAATTCGGGGCTCAACCCCGTCCGGCCGTTGA
ASV5	TACGGAGGATCCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGAGCGTAGGTGGATTGTTAAGTCAGTTGTGAAAGTTTGCGGCTCAACCGTAAAATTGCAGTTGA
ASV6	TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAGCCGTTGA
ASV7	TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGATGCCAAGTCAGCGGTAAAAAAGCGGTGCTCAACGCCGTCGAGCCGTTGA
ASV8	TACGGAGGATTCAAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGTTCGATAAGTTAGAGGTGAAATCCCGGGGCTCAACTCCGGCACTGCCTCTGA
ASV9	TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGATCGTTAAGTCAGTGGTCAAATTGAGGGGCTCAACCCCTTCCCGCCATTGAA
ASV10	TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGATGTCAAGTCAGCGGTAAAATTGTGGAGCTCAACTCCATCGAGCCGTTGAA
ASV11	TACGTAGGTGGCAAGCGTTGTCCGGATTTATTGGGCGTAAAGCGAGTGCAGGCGGTTCAATAAGTCTGATGTGAAAGCCTTCGGCTCAACCGGAGAATTGCATCAGA
ASV12	TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGCCTGCCAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTACAGCCGTTGA
ASV13	TACGTAGGTGGCAAGCGTTATCCGGATTTATTGGGCGTAAAGGGAACGCAGGCGGTCTTTTAAGTCTGATGTGAAAGCCTTCGGCTTAACCGGAGTAGTGCATTGGA
ASV14	TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGGATGCCAAGTCAGCGGTAAAAATGCGGTGCTCAACGCCGTCGAGCCGTTGA
ASV15	TACGTAGGGGGCAAGCGTTATCCGGATTTACTGGGTGTAAAGGGAGCGTAGACGGCAGCGCAAGTCTGGAGTGAAATGCCGGGGCCCAACCCCGGAACTGCTTTG
ASV16	TACGTAGGTGGCAAGCGTTATCCGGAATTATTGGGCGTAAAGAGCGCGCAGGTGGTTAATTAAGTCTGATGTGAAAGCCCACGGCTTAACCGTGGAGGGTCATTGGA
ASV17	TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGCAGGCGGGATGCCAAGTCAGCGGTCAAATTTCGGGGCTCAACCCCGACCTGCCGTTGA
ASV18	TACGGAGGATGCGAGCGTTATCCGGATTTATTGGGTTTAAAGGGTGCGTAGGCGGTCCGTTAAGTCAGCGGTAAAATTGCGGGGCTCAACCCCGTCGAGCCGTTGA
ASV19	TACGTAGGGAGCGAGCGTTATCCGGATTTATTGGGTGTAAAGGGTGCGTAGGCGGTAATACAGGTCTTTGGTATAAGCCCGAAGCTTAACTTCGGTAAGCCAGAGAA

IonTorrent or ITS data with DADA2



ASV

Outline

- Main parts of ITS and IonTorrent analysis with DADA2
- What are the differences compared to Illumina data
- Things to take into account

Things to note

- Use Cutadapt to remove the adapter/primer sequences
- When filtering and trimming your reads, use "Remove reads which are shorter than this" instead of truncation
  - If the read length varies
- IonTorrent:
  - Use the parameter IonTorrent in the tool Sample inference
    - Uses specific denoising algorithm options recommended for IonTorrent data
    - HOMOPOLYMER_GAP_PENALTY = -1
    - BANDSIZE= 32
- ITS
  - Download the UNITE reference files from the website and give as input
  - But: see next slide